Calculus and Linear Algebra

Infinite Limits and Limits at Infinity

Lecture 14

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Lecture outline

- Infinite limits at a finite point.
- Rigorous definitions using an arbitrary level M.
- One-sided infinite limits and vertical asymptotes.
- Finite and infinite limits as $x \to \pm\infty$.
- Horizontal asymptotes.
- Comparison principles and dominant terms.
- Indeterminate forms and algebraic techniques.

Main goal

Complete the basic language of limits before moving to continuity.

Four different questions

Limits can involve a finite or infinite input and a finite or infinite output:

$$\boxed{\lim_{x \to a} f(x) = L} \qquad \boxed{\lim_{x \to a} f(x) = \pm\infty}$$

$$\boxed{\lim_{x \to \pm\infty} f(x) = L} \qquad \boxed{\lim_{x \to \pm\infty} f(x) = \pm\infty.}$$

Each statement has a precise meaning. The symbol ∞ describes unbounded behaviour; it is not a real number.

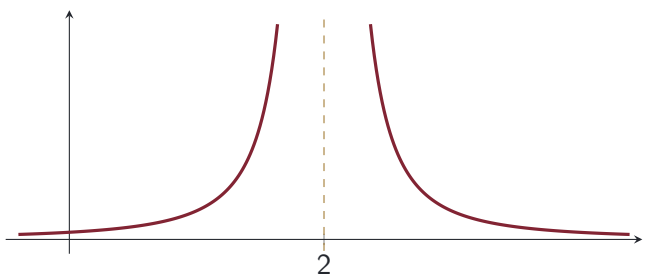
Infinite limit at a finite point: intuition

Consider

$$f(x) = \frac{1}{(x-2)^2}.$$

As x approaches 2 from either side, the function becomes arbitrarily large and positive:

$$\boxed{\lim_{x \to 2} \frac{1}{(x-2)^2} = +\infty.}$$



Rigorous definition: $+\infty$ at a finite point

Definition We write

$$\boxed{\lim_{x \to a} f(x) = +\infty}$$

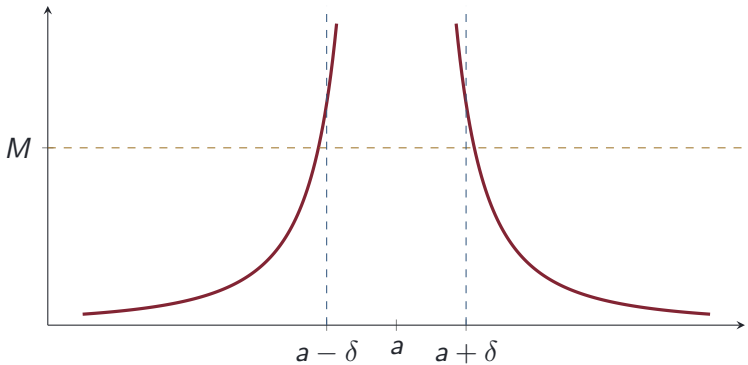
if

$$\boxed{\forall M > 0 \ \exists \delta > 0 \text{ such that } 0 < |x - a| < \delta \Rightarrow f(x) > M.}$$

Meaning

No matter how high a horizontal level M is chosen, sufficiently close to a the graph lies above that level.

The M–δ picture



Choosing a larger M generally forces us to choose a smaller δ.

Rigorous definition: $-\infty$ at a finite point

We write

$$\boxed{\lim_{x \to a} f(x) = -\infty}$$

if

$$\boxed{\forall M > 0 \; \exists \delta > 0 \text{ such that } 0 < |x - a| < \delta \Rightarrow f(x) < -M.}$$

The quantity M is positive; the condition $f(x) < -M$ forces the values to become arbitrarily negative.

One-sided infinite limits

For

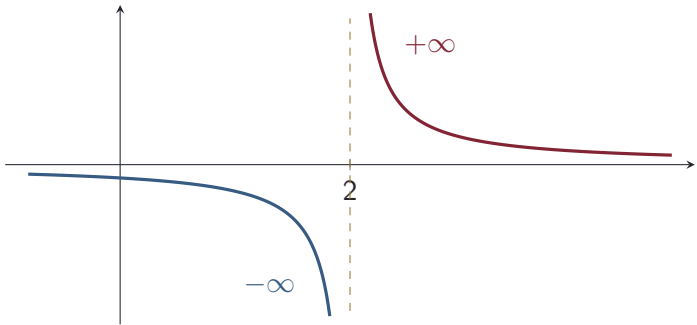
$$f(x) = \frac{1}{x-2},$$

the two sides behave differently:

$$\boxed{\lim_{x \to 2^-} \frac{1}{x-2} = -\infty,} \qquad \boxed{\lim_{x \to 2^+} \frac{1}{x-2} = +\infty.}$$

Therefore there is no common two-sided infinite limit with the same sign.

One-sided behaviour: see the two branches



The sign of the denominator is crucial for one-sided infinite limits.

Vertical asymptotes

Definition The line

$$\boxed{x = a}$$

is a **vertical asymptote** of f if at least one of the one-sided limits at a is $+\infty$ or $-\infty$.
Examples:

$$f(x) = \frac{1}{x-2} \quad \Rightarrow \quad x = 2,$$

$$g(x) = \frac{1}{x^2-1} \quad \Rightarrow \quad x = -1,\ x = 1.$$

Domain and limits work together

A zero of the denominator is a candidate for a vertical asymptote, but simplification may instead produce a removable hole.

Hole or vertical asymptote?

Compare

$$f(x) = \frac{x^2 - 1}{x - 1} \qquad \text{and} \qquad g(x) = \frac{x + 1}{x - 1}.$$

At $x = 1$:

$$f(x) = x + 1 \quad (x \neq 1),$$

so

$$\lim_{x \to 1} f(x) = 2$$

and there is a hole, not a vertical asymptote. For g,

$$\lim_{x \to 1^-} g(x) = -\infty, \qquad \lim_{x \to 1^+} g(x) = +\infty.$$

Thus $x = 1$ is a vertical asymptote of g.

Finite limits as $x \to +\infty$

Consider

$$f(x) = 1 + \frac{1}{x}.$$

As x becomes arbitrarily large,

$$f(x) \to 1.$$

We write

$$\boxed{\lim_{x \to +\infty} \left(1 + \frac{1}{x}\right) = 1.}$$

This describes the long-run behaviour of the function to the right.

Rigorous definition: finite limit at $+\infty$

Definition We write

$$\boxed{\lim_{x \to +\infty} f(x) = L}$$

if

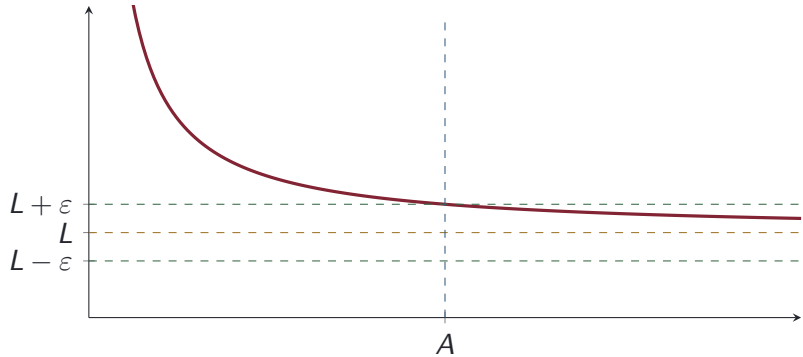
$$\boxed{\forall \varepsilon > 0 \ \exists A > 0 \text{ such that } x > A \Rightarrow |f(x) - L| < \varepsilon.}$$

For $x \to -\infty$, replace $x > A$ by $x < -A$.

Compare with ε–δ

Now the input is not required to be close to a finite point; it is required to be sufficiently far along the real axis.

The ε–A picture



Beyond a sufficiently large A, the graph remains inside the horizontal ε-band around L.

Horizontal asymptotes

Definition If

$$\lim_{x \to +\infty} f(x) = L \quad \text{or} \quad \lim_{x \to -\infty} f(x) = L,$$

then

$$\boxed{y = L}$$

is a **horizontal asymptote** on the corresponding side. A function may have different horizontal asymptotes as $x \to +\infty$ and $x \to -\infty$.

Infinite limits as $x \to +\infty$

Definition We write

$$\boxed{\lim_{x \to +\infty} f(x) = +\infty}$$

if

$$\boxed{\forall M > 0 \; \exists A > 0 \text{ such that } x > A \Rightarrow f(x) > M.}$$

Similarly,

$$\lim_{x \to +\infty} f(x) = -\infty$$

means that for every $M > 0$, eventually $f(x) < -M$.

Examples at infinity

For positive integer powers,

$$\lim_{x \to +\infty} x^n = +\infty.$$

As $x \to -\infty$:

$$\lim_{x \to -\infty} x^{2n} = +\infty,$$

$$\lim_{x \to -\infty} x^{2n+1} = -\infty.$$

Also,

$$\lim_{x \to \pm\infty} \frac{1}{x^n} = 0.$$

These elementary behaviours are the building blocks for polynomial and rational limits.

The four basic limit types

Input	Output	Typical example
$x \to a$	$L \in \mathbb{R}$	$\frac{x^2-1}{x-1} \to 2$ at $a = 1$
$x \to a$	$\pm\infty$	$\frac{1}{(x-a)^2} \to +\infty$
$x \to \pm\infty$	$L \in \mathbb{R}$	$1 + \frac{1}{x} \to 1$
$x \to \pm\infty$	$\pm\infty$	$x^3 \to \pm\infty$

Knowing which type we are studying determines the appropriate definition and graphical interpretation.

Infinity is not ordinary arithmetic

Expressions such as

$$\frac{\infty}{\infty}, \qquad \infty - \infty, \qquad 0 \cdot \infty$$

are not numbers and cannot be manipulated by ordinary arithmetic. They are

indeterminate forms: different functions producing the same symbolic form may have completely different limits.

Why ∞/∞ is indeterminate

As $x \to +\infty$:

$$\frac{x}{x} \to 1, \qquad \frac{x}{x^2} \to 0, \qquad \frac{x^2}{x} \to +\infty.$$

All three direct symbolic substitutions look like

$$\frac{\infty}{\infty},$$

but the limits are completely different.

Conclusion

An indeterminate form tells us to transform or compare the functions; it is never the final answer.

Comparison principle for infinite limits

Theorem Suppose that, for x sufficiently close to a,

$$f(x) \geq g(x),$$

and

$$\lim_{x \to a} g(x) = +\infty.$$

Then

$$\boxed{\lim_{x \to a} f(x) = +\infty.}$$

Analogous statements hold for $-\infty$ and for limits as $x \to \pm\infty$.

Squeeze theorem at infinity

The squeeze theorem also works when the input tends to infinity. If, for all sufficiently large x,

$$g(x) \leq f(x) \leq h(x)$$

and

$$\lim_{x \to +\infty} g(x) = \lim_{x \to +\infty} h(x) = L,$$

then

$$\boxed{\lim_{x \to +\infty} f(x) = L.}$$

The same principle applies for $x \to -\infty$.

Dominant terms in a polynomial

For large $|x|$, the highest power dominates a polynomial. For example,

$$p(x) = 3x^4 - 2x^2 + 7x - 5.$$

Factor x^4:

$$p(x) = x^4 \left(3 - \frac{2}{x^2} + \frac{7}{x^3} - \frac{5}{x^4}\right).$$

The bracket tends to 3, so

$$\boxed{p(x) \sim 3x^4 \text{ for large } |x|.}$$

Hence

$$\lim_{x \to \pm\infty} p(x) = +\infty.$$

Rational functions at infinity: compare degrees

For

$$f(x) = \frac{P_m(x)}{Q_n(x)},$$

where m and n are the polynomial degrees:

- $m < n$: typically $f(x) \to 0$;
- $m = n$: the limit is the ratio of leading coefficients;
- $m > n$: the magnitude is generally unbounded or follows a polynomial asymptotic behaviour.

Reason

Divide numerator and denominator by the highest relevant power of x.

Degree of numerator smaller than denominator

Compute

$$\lim_{x\to+\infty} \frac{3x+1}{x^2+4}.$$

Divide by x^2:

$$\frac{\frac{3}{x}+\frac{1}{x^2}}{1+\frac{4}{x^2}}.$$

Since $1/x \to 0$ and $1/x^2 \to 0$,

$$\boxed{\lim_{x\to+\infty} \frac{3x+1}{x^2+4} = 0.}$$

Thus $y = 0$ is a horizontal asymptote.

Same degree: ratio of leading coefficients

Compute

$$\lim_{x\to-\infty} \frac{2x^2-3x+1}{5x^2+x-4}.$$

Divide by x^2:

$$\frac{2-\frac{3}{x}+\frac{1}{x^2}}{5+\frac{1}{x}-\frac{4}{x^2}}.$$

Therefore

$$\boxed{\lim_{x\to-\infty} \frac{2x^2-3x+1}{5x^2+x-4} = \frac{2}{5}.}$$

The same limit holds as $x \to +\infty$.

Numerator degree larger than denominator

Consider

$$\frac{x^3+1}{x^2+1}.$$

Divide by x^2:

$$\frac{x+1/x^2}{1+1/x^2}.$$

Hence the function behaves like x for large $|x|$:

$$\lim_{x\to+\infty}\frac{x^3+1}{x^2+1}=+\infty,$$

$$\lim_{x\to-\infty}\frac{x^3+1}{x^2+1}=-\infty.$$

Later, polynomial division will also identify possible oblique asymptotes.

A finite-point 0/0 form: factor first

Compute

$$\lim_{x \to 3} \frac{x^2 - 9}{x - 3}.$$

Factor:

$$x^2 - 9 = (x - 3)(x + 3).$$

For $x \neq 3$,

$$\frac{x^2 - 9}{x - 3} = x + 3.$$

Therefore

$$\boxed{\lim_{x \to 3} \frac{x^2 - 9}{x - 3} = 6.}$$

Rationalization at a finite point

Compute

$$\lim_{x\to 0} \frac{\sqrt{x+1}-1}{x}.$$

Using the conjugate,

$$\frac{\sqrt{x+1}-1}{x} = \frac{1}{\sqrt{x+1}+1} \quad (x \neq 0).$$

Thus

$$\boxed{\lim_{x\to 0} \frac{\sqrt{x+1}-1}{x} = \frac{1}{2}.}$$

The original $0/0$ form disappears after the algebraic transformation.

Rationalization at infinity

Compute

$$\lim_{x\to+\infty}\left(\sqrt{x^2+x}-x\right).$$

The direct symbolic form is $\infty-\infty$. Multiply by the conjugate:

$$\sqrt{x^2+x}-x=\frac{x}{\sqrt{x^2+x}+x}.$$

For $x>0$, divide numerator and denominator by x:

$$\frac{1}{\sqrt{1+1/x}+1}.$$

Hence

$$\boxed{\lim_{x\to+\infty}(\sqrt{x^2+x}-x)=\frac{1}{2}.}$$

One-sided infinite limits: use signs

Consider

$$f(x) = \frac{1}{(x-1)(x+2)}.$$

Potential vertical asymptotes are

$$x = -2, \qquad x = 1.$$

To determine the sign of each infinite limit, study the sign of the denominator on each side.

Interval	$(-\infty, -2)$	$(-2, 1)$	$(1, +\infty)$
$(x-1)(x+2)$	$+$	$-$	$+$
$f(x)$	$+$	$-$	$+$

One-sided infinite limits: conclusions

For

$$f(x) = \frac{1}{(x-1)(x+2)},$$

we obtain

$$\lim_{x \to -2^-} f(x) = +\infty, \qquad \lim_{x \to -2^+} f(x) = -\infty,$$

$$\lim_{x \to 1^-} f(x) = -\infty, \qquad \lim_{x \to 1^+} f(x) = +\infty.$$

Thus both $x = -2$ and $x = 1$ are vertical asymptotes.

Common mistakes with infinite limits

- Treating ∞ as an ordinary real number.
- Writing $1/0 = \infty$ without checking the side and the sign.
- Ignoring left-hand and right-hand behaviour at a denominator zero.
- Declaring every denominator zero a vertical asymptote without simplifying first.
- Using the ratio of leading coefficients when numerator and denominator degrees are different.
- Forgetting that $\sqrt{x^2} = |x|$, especially as $x \to -\infty$.
- Treating an indeterminate form as a final result.

A subtle point: $\sqrt{x^2} = |x|$

As $x \to +\infty$,

$$\sqrt{x^2} = x.$$

But as $x \to -\infty$,

$$\sqrt{x^2} = |x| = -x.$$

Therefore sign-sensitive simplifications involving square roots must be handled carefully.

Example

$$\frac{\sqrt{x^2}}{x} = \frac{|x|}{x} \to \begin{cases} 1, & x \to +\infty, \\ -1, & x \to -\infty. \end{cases}$$

Guided exercise A

Compute

$$\lim_{x \to 2^-} \frac{3}{x-2}, \qquad \lim_{x \to 2^+} \frac{3}{x-2}.$$

Then state the vertical asymptote and describe the two branches near it.

Guided exercise A: solution

For $x < 2$ close to 2, $x - 2$ is a small negative number, so

$$\boxed{\lim_{x \to 2^-} \frac{3}{x-2} = -\infty.}$$

For $x > 2$ close to 2, the denominator is a small positive number, so

$$\boxed{\lim_{x \to 2^+} \frac{3}{x-2} = +\infty.}$$

Therefore

$$\boxed{x = 2}$$

is a vertical asymptote.

Guided exercise B

Compute

$$\lim_{x \to +\infty} \frac{4x^2 - x + 7}{2x^2 + 3x - 1}$$

and

$$\lim_{x \to +\infty} \frac{5x - 2}{x^3 + 1}.$$

Use the degree comparison and then justify it by dividing by the appropriate power of x.

Guided exercise B: solution

For the first quotient, numerator and denominator have the same degree:

$$\boxed{\lim_{x\to+\infty} \frac{4x^2 - x + 7}{2x^2 + 3x - 1} = \frac{4}{2} = 2.}$$

For the second, the denominator has higher degree:

$$\frac{5/x^2 - 2/x^3}{1 + 1/x^3} \to 0.$$

Thus

$$\boxed{\lim_{x\to+\infty} \frac{5x - 2}{x^3 + 1} = 0.}$$

Independent practice

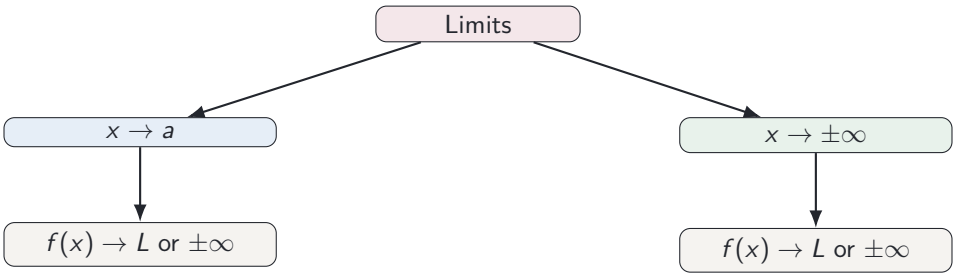
Compute the following limits.

1. $\lim_{x \to 1^-} \frac{1}{x-1}$ and $\lim_{x \to 1^+} \frac{1}{x-1}$.
2. $\lim_{x \to \pm\infty} \frac{2x+1}{x^2+1}$.
3. $\lim_{x \to +\infty} \frac{3x^3 - x}{x^3 + 4}$.
4. $\lim_{x \to -\infty} \frac{x^3 + 2}{x^2 + 1}$.
5. $\lim_{x \to 4} \frac{\sqrt{x} - 2}{x - 4}$.

Independent practice: answers

1. $-\infty$ from the left, $+\infty$ from the right.
2. 0 in both directions.
3. 3.
4. $-\infty$.
5. Rationalize: $\frac{1}{\sqrt{x}+2} \to \frac{1}{4}$.

The conceptual map of limits



We now have the definitions, graphical meanings and first computational tools for all four cases.

Next: continuity

The next natural question is:

When does the limiting value equal the actual function value?

This leads to

$$\lim_{x \to a} f(x) = f(a),$$

which is the definition of continuity at a. In the next lecture we can study continuity, discontinuities and the main theorems for continuous functions without having to rush the theory of limits.